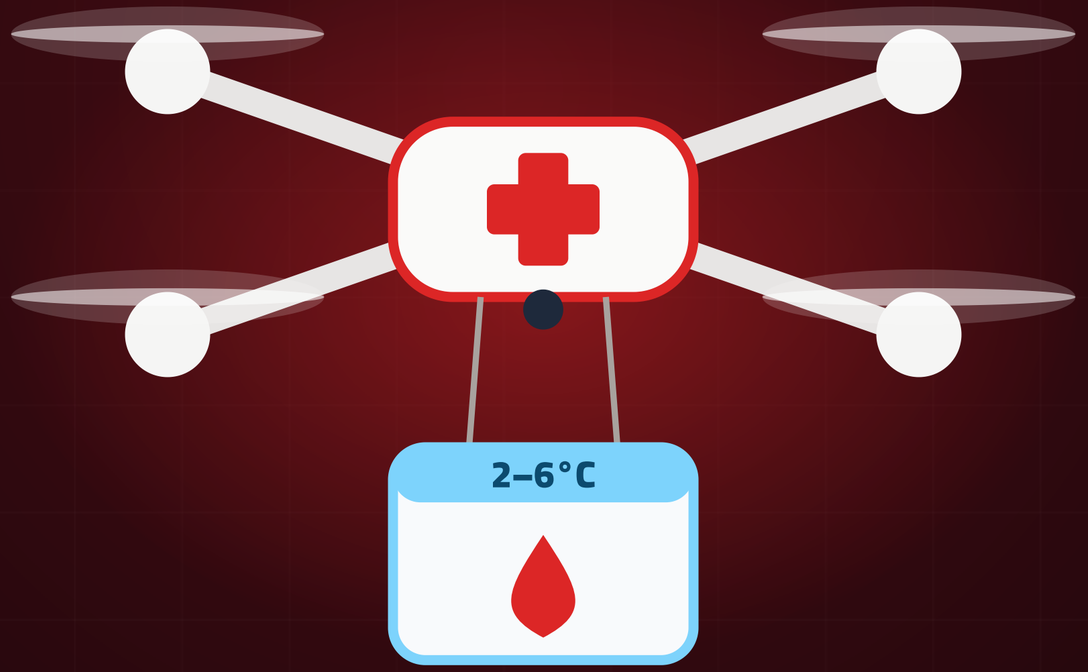
DAMI

Blood that arrives… before it's too late

An AI platform and medical drones for managing and delivering blood in emergencies

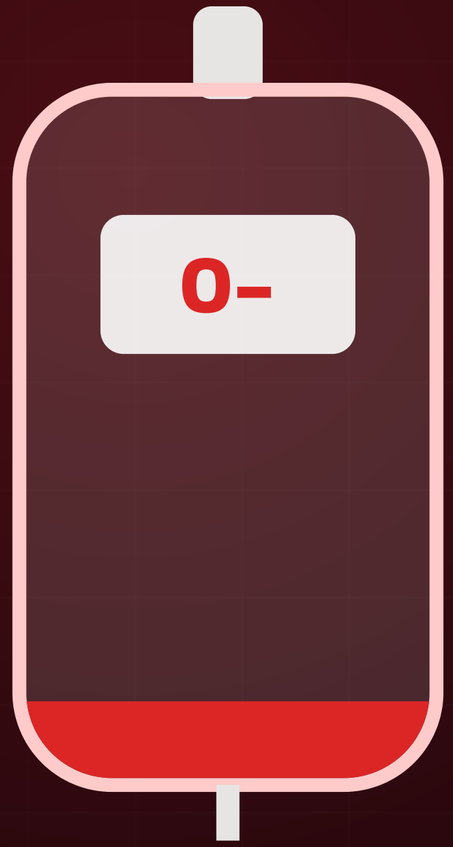


Team NEXORA · AI & Data Science

30–40%

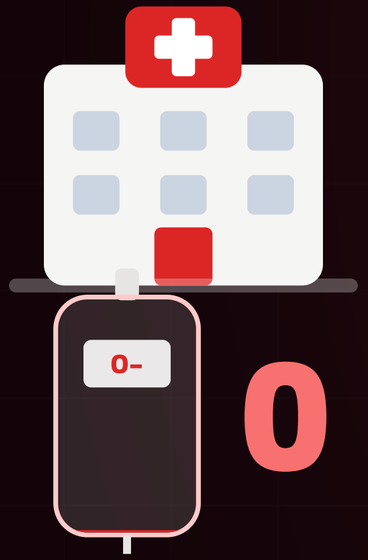
of trauma deaths are caused by bleeding

and every minute of delay raises the risk

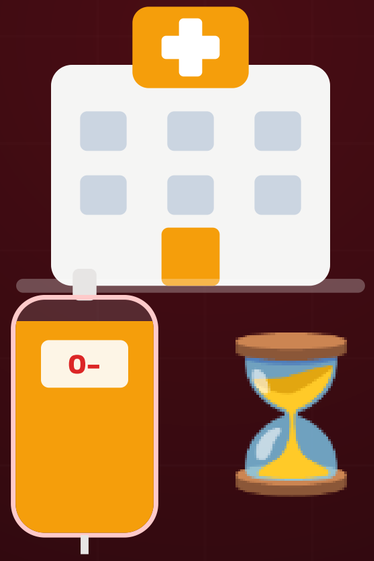


Source: Kauvar et al., Journal of Trauma, 2006

Shortage here... waste there



One hospital needs O– and has none



while units expire at another

5–7 days
platelet shelf life

Traffic
delays emergency transport

Generic drives
miss the missing type

Why does blood run short?



Seasonal donation

drops in summer, Ramadan and exams



Short shelf life

platelets 5–7 days, red cells 35–42

Rare types

O– always needed, rarely available



Uneven distribution

stock in the center, outskirts wait



Sudden emergencies

one accident uses days of stock

Manual coordination

phone calls, no live view

The solution: DAMI



Forecast & balance

before shortage or waste



Call the donor

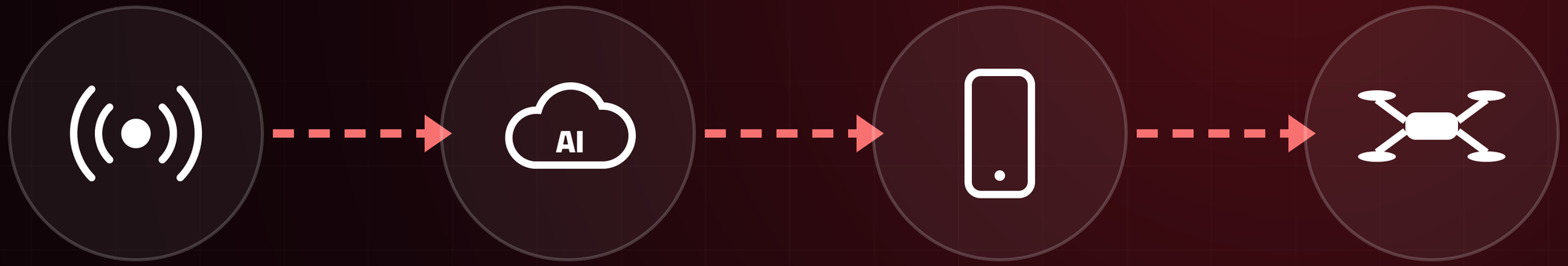
the right one, nearby



Blood drone

in minutes

How the system works



Blood-bank sensors

temperature + live stock

DAMI AI cloud

forecast + best source

Hospital app

one-tap emergency

Cooled drone

GPS + live temperature

The system in depth: 8 smart modules

🏷️

Smart inventory

RFID per unit with expiry

🧠

Demand forecasting

per type & hospital, 14 days

⚖️

Smart balancing

move units before they expire

🔍

Matching engine

compatible type, nearest source

🚁

Fleet management

safe route, licensed pilot approval

🌡️

Cold chain

temperature every second, alarms

🫂

Donor engine

targeted call-up for missing type

🏛️

National dashboard

live view for the Ministry

Emergency request: from tap to arrival

0 s
Hospital requests 6 units O−

5 s
AI checks stock & compatibility

10 s
Best source: drone or car

1 min
Pilot approval, box packed

≈ 9 min
Flight with live temperature & GPS

✓
Delivered, donors called to restock

Times from the simulation for a typical distance in Amman

The blood ambulance drone

2–6 °C box

temperature every second

4–6 units

per flight

Minutes

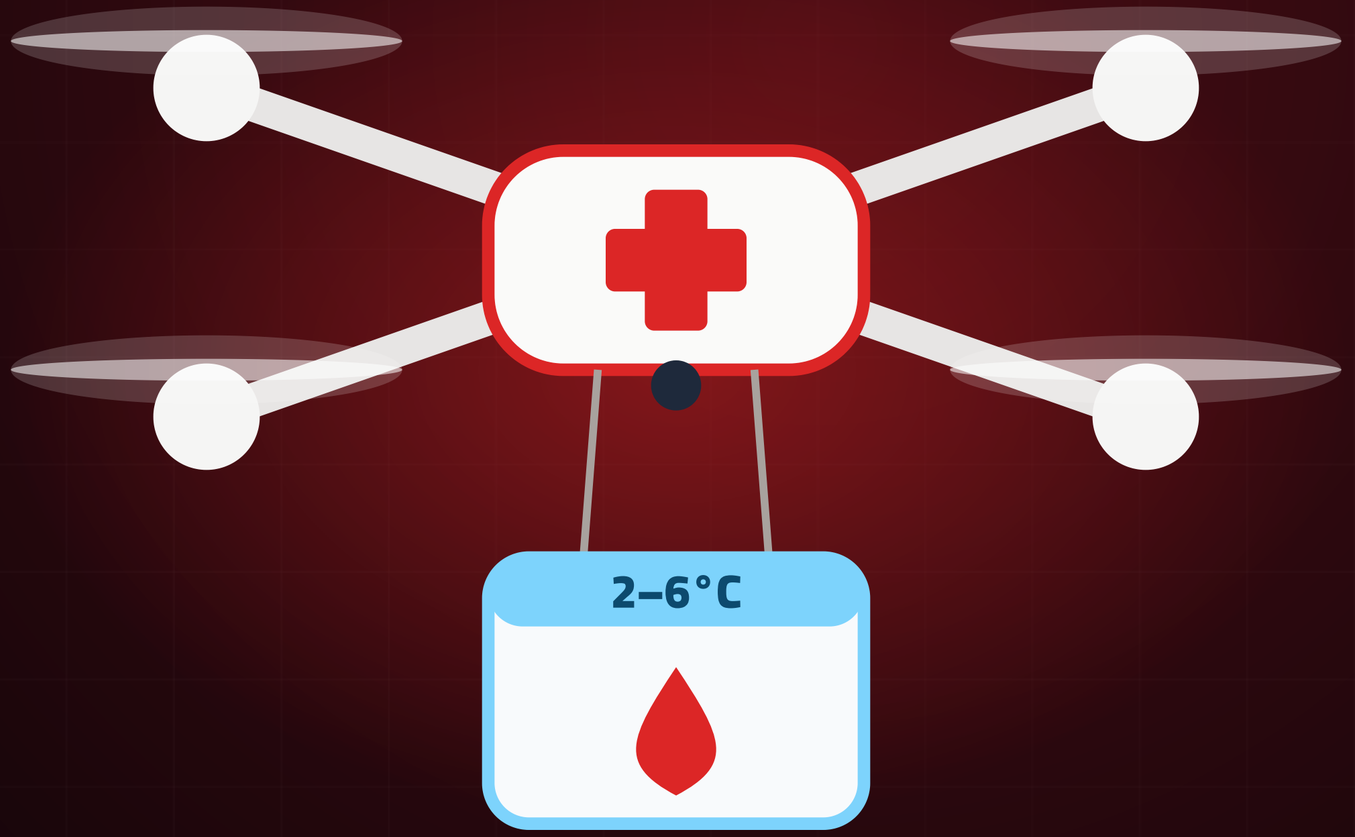
not half an hour

Live tracking

GPS, safe route

Safety

parachute, auto-return



Sensors & the cold chain

Blood-bank fridge

temperature & live stock sensor

Every blood unit

ID tag and expiry date

Drone box

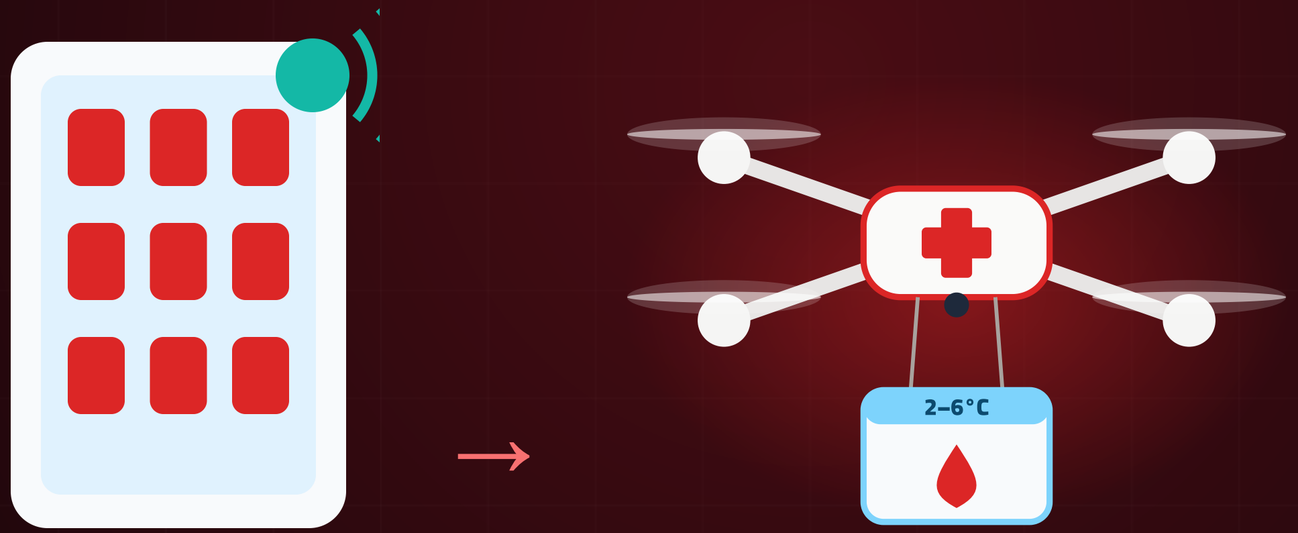
2–6°C every second + GPS

Instant alarm

backup cooling, hospital notified

Full log

temperature report with every shipment



Matching engine: always the right type

O− gives to everyone

AB+ receives from everyone

If a type runs out, it proposes a compatible alternative instantly



The smart donor

✅ **Eligible**

last donation 90+ days ago

📍 **Nearby**

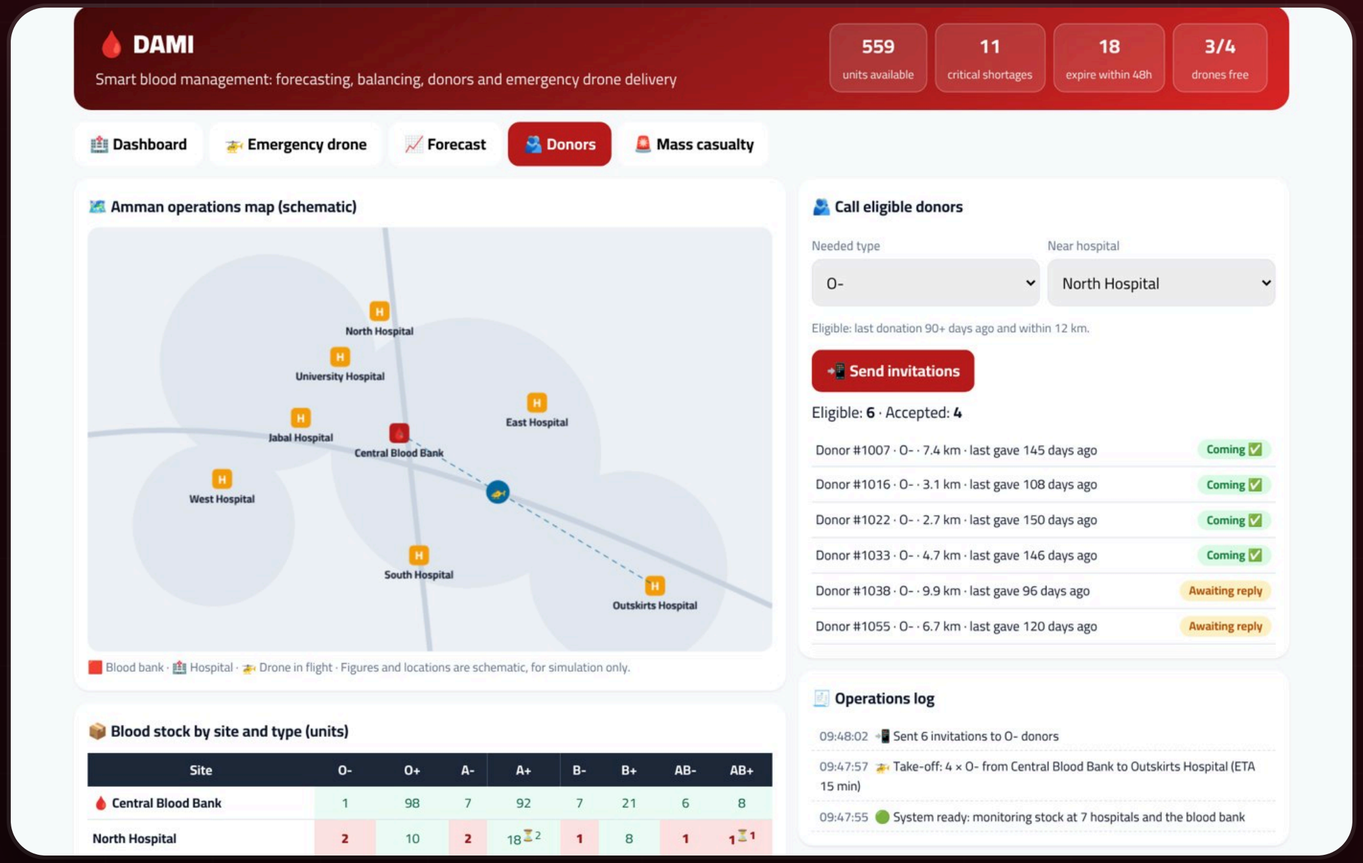
within 12 km of the hospital

🎯 **Targeted**

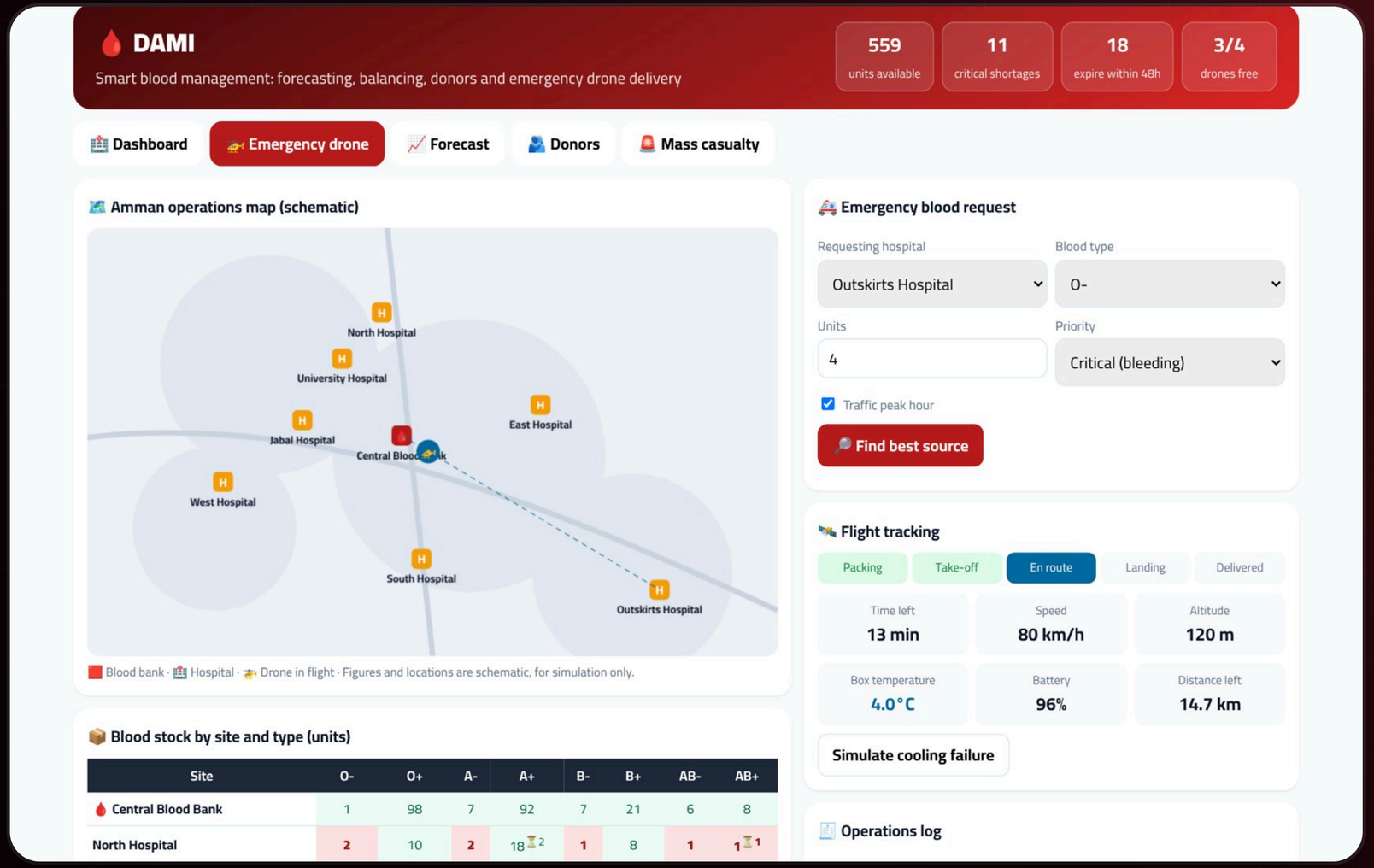
only for the missing type

❤️ **Visible impact**

knows where their unit went



The prototype works today



① Picks the best source

② Compares drone vs car

③ Alarms instantly if temperature rises

Proven worldwide

67%

drop in blood wastage after drone delivery in Rwanda

79 min

faster delivery

🇯🇴 **No similar system in Jordan per our preliminary research**

Source: Nisingizwe et al., The Lancet Global Health, 2022

Who pays?



JOD 6,000

hospital subscription / year



JOD 90

per emergency flight



JOD 40,000

national dashboard / year



JOD 10,000

donor-app sponsorship / year

72% gross margin in year 3

The plan

Now

Prototype ✓

6–12 months

3 hospitals

12–24 months

First licensed drone route

24–36 months

National dashboard

Our ask



Seed funding

≈ JOD 90,000



Pilot partnership

blood bank + hospital



Permit support

Civil Aviation Regulatory Commission

Team NEXORA · AI & Data Science

Team NEXORA

AI & Data Science students

Abdulrahman Bilal Aburumman
Team Leader

Marah Yousef Al-Qaisi

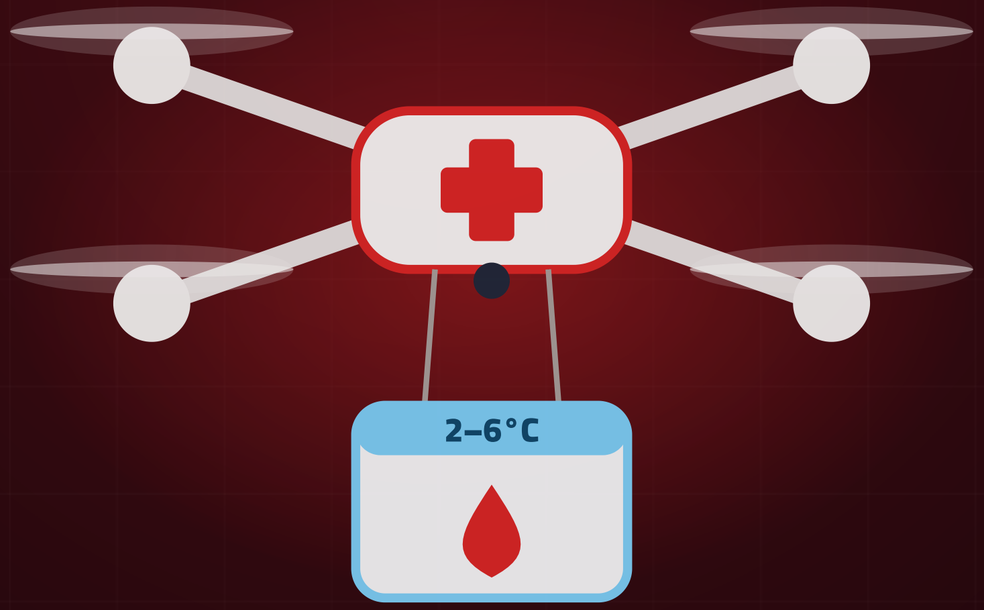
Nadeen Mohammad Al-Smadi

Darya Ali Al-Shaqaqha

Abdulmonem Fadi Qadan

Because blood that arrives late...

doesn't arrive.



DAMI · NEXORA